

Countability

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Let pred be the function $\mathbb{N} \rightarrow \mathcal{P}\mathbb{N}$ defined recursively by

$$\text{pred } 0 = \emptyset$$

$$\text{pred}(\text{suc } n) = \text{pred } n \cup \{n\}$$

We write $n < m$ as shorthand for ' $n \in \text{pred } m$ ', and $n \leq m$ as shorthand for ' $n < m$ or $m = m$ ', so the recursion clauses can be written as follows:

$$n < 0 \text{ never}$$

$$n < \text{suc } m \text{ iff } n \leq m$$

Some facts about order

Useful facts

For all k, n, m :

1. $0 \leq n$
2. If $\text{succ } n < m$ then $n < m$.
3. If $k < n$ and $n < m$ then $k < m$ (And thus if $k \leq n$ and $n \leq m$ then $k \leq m$.)
4. Not $n < n$.
5. Not $n < m$ and $m < n$. (And thus if $n \leq m$ and $m \leq n$, $n = m$.)
6. Either $n < m$ or $m < n$ or $n = m$ (And thus either $n \leq m$ or $m \leq n$.)
7. $n \leq m$ iff $m = n + k$ for some k .

These can all be proved by straightforward inductions.

Example

For example, here's the proof that $k < n$ and $n < m$ then $k < m$. By induction on m , generalizing over k and n .

Base case: trivial since it can't happen that $k < n$ and $m < 0$.

Induction step: suppose that whenever $k < n$ and $n < m$, $k < m$, and that for a certain k and n , $k < n$ and $n < \text{succ } m$. Then either $n < m$, in which case $k < m$ by the IH, or else $n = m$, in which case $k < m$ by substitution; either way, we have $k < \text{succ } m$ by the recursion clause for $<$.

Another example

Proof by induction that every n is such that there is no m for which $n < m$ and $m < n$.

Base case: trivial since there is no m for which $m < 0$.

Induction step: Suppose that there is no m for which $n < m$ and $m < n$, and that $m < \text{succ } n$. So either $m < n$ or $m = n$. If $m < n$, then not $n < m$ by the IH; if $m = n$, then not $n < m$ by substitution. So either way, not $\text{succ } n < m$ by fact 2.

One more useful fact

Least Number Principle

For every $X \subseteq \mathbb{N}$, either $X = \emptyset$ or there exists $n \in X$ such that $n \leq m$ for all $m \in X$.

This is equivalent to the claim that $<$ is *well-founded*:

Definition

Where R is a binary relation on A , R is *well-founded* iff for every $X \subseteq A$, either $X = \emptyset$ or there exists $x \in X$ such that there is no $y \in X$ for which Ryx .

This can be used to justify a different kind of proof by induction, so called “strong induction”. If we want to prove that every number has property ϕ , we can do so by showing that for any n , if $\phi(m)$ for all $m < n$, then $\phi(n)$. (In other words: there is no least non- ϕ number; so by the Least Number Principle, there is no non- ϕ number.)

Proving the Least Number Principle

Suppose that X has no least element. First we will prove by induction that for all n , $X \cap \text{pred } n = \emptyset$.

Base case: trivial since $\text{pred } 0 = \emptyset$.

Induction hypothesis: suppose for contradiction that $X \cup \text{pred } n = \emptyset$ and $m \in X \cup \text{pred}(\text{succ } n)$. Then we must have $m \in \text{pred}(\text{succ } n) \setminus \text{pred } n$, hence $m = n$: but then n is a least element of X .

This implies that $X = \emptyset$. For suppose $n \in X$; then since $n \in \text{pred}(\text{succ } n)$, we would have $n \in X \cap \text{pred}(\text{succ } n)$.

For lists there are two analogues of $\leq$, the *initial segment* and *final segment* relations, and two analogues of $<$, the *proper initial segment* and *proper final segment* relations.

Everything we have said about $<$ applies *mutatis mutandis* to these principles, except that we don't have the analogue of the 'connectedness' fact: we can have two lists neither of which is an initial/final segment of the other.

Definition

$n \in \mathbb{N}$ is the *size* of $A := A \sim \text{pred } n$.

In an earlier lecture, we defined ‘ A is finite’ to mean ‘ $A \in \mathcal{P}_{\text{fin}} A$ ’, where $\mathcal{P}_{\text{fin}} A$ (the set of finite subsets of A) is defined as the closure of $\{\emptyset\}$ under the operation of adding one element of A .

We could equivalently have defined finitude in terms of the natural numbers:

Fact

A is finite iff $A \sim \text{pred } n$ for some $n \in \mathbb{N}$.

For the left to right direction, we prove by induction that for every $B \in \mathcal{P}_{fin}A$ there is some n such that $B \sim \text{pred } n$. For $\emptyset$ it's 0; and when B' is the result of adding one element to B and $B \sim n$, $B' \sim \text{succ } n$.

For the right to left direction, it suffices to note that $\text{pred } n$ is finite for every n (since $\emptyset$ is finite, and adding one element to a finite set always produces a finite set). Then we have to show that if $A \sim B$ and A is finite, B is finite.

Given that a finite set is one that has size n for some n , the following can readily be established by numerical induction:

Finitude Facts

1. If A and B are finite, $A \cup B$ is finite.
2. If A and B are finite, $A \times B$ is finite.
3. If A is a finite set of finite sets, $\bigcup A$ is finite.
4. If A is finite, $\mathcal{P}A$ is finite.
5. If A and B are finite, A^B is finite.

Definition

Set A is *countable* $:= A \lesssim \mathbb{N}$. (I.e.: there is an injection from A to $\mathbb{N}$.)

Don't confuse with:

Definition

Set A is *countably infinite* $:= A \sim \mathbb{N}$ (there is a bijection from A to $\mathbb{N}$.)

Fact

A set is countable iff it is either finite or countably infinite.

To prove this, we show that every subset of $\mathbb{N}$ that is not a subset of $\text{pred } n$ for any n is the same size as $\mathbb{N}$. We define an injection:

$$\begin{aligned} f0 &= \min X \\ f \text{ suc } n &= \min(X \setminus (fn \cup \text{pred } fn)) \end{aligned}$$

This is an injection from $\mathbb{N}$ to X .

Fact

If A is finite and $\neq \emptyset$, then A^* is countably infinite.

(Note that $\emptyset^* = \{\epsilon\}$ which is not countably infinite, though it is countable.)

Intuitively, this is true because we can list all the members of A^* in alphabetical order.

Strings over a finite alphabet

To define an injective function f from A^* to $\mathbb{N}$, let n be the size of A , and let g be a bijection from A to $\text{pred } n$. Let $g^+a = \text{succ } ga$ for every $a \in A$.

We define $f : A^* \rightarrow \mathbb{N}$ inductively as follows.

$$\begin{aligned}f[] &= 0 \\f(a : s) &= g^+a + n \times fs\end{aligned}$$

This turns out to be injective (in fact it's a bijection). To prove it's injective, we rely on the following well-known arithmetical fact:

Division Theorem

If $qn + r = q'n + r'$, where $r < n$ and $r' < n$, then $q = q'$ and $r = r'$.

Strings over a finite alphabet

We prove by induction (on t) that whenever $fs = ft$, $s = t$.

Base case: suppose $fs = f[] = 0$. Then it can't be that s is $(a : s')$ for some a and s' , since then we'd have $fs = \text{suc}(ga + nfs')$, and zero isn't a successor. So it must be that $s = []$.

Induction step: suppose that s is such that whenever $fs = ft$, $s = t$, and suppose $f(a : s) = ft$. It can't be that $t = []$ since $f(a : s) \neq 0$, so we must have $t = b : t'$ for some b, t' . So we have $\text{suc}(nfs + ga) = \text{suc}(ngt' + gb)$, and hence $nfs + ga = nft' + gb$. Since both $ga < n$ and $gb < n$, the division theorem implies $fs = ft'$ and $ga = gb$. But then $s = t'$ by the induction hypothesis, and $a = b$ by the injectivity of b , so $t = (b : t') = (a : s)$.

Proving the division theorem

Coding lists of lists as lists

Suppose a finite set $B = A \cup \{c\}$. Then we can define an injection from $(A^*)^*$ to B^* by using c as a ‘comma’ to join any list of A -lists into one big B -list.

We define $f : A^{**} \rightarrow B^*$ recursively as follows.

$$\begin{aligned} f[] &= [] \\ f(s : j) &= \begin{cases} s & \text{if } j = [] \text{ and } s \neq [] \\ s \oplus (c : fj) & \text{otherwise} \end{cases} \end{aligned}$$

To show that this is injective, we can define a ‘decoding’ function $g : B^* \rightarrow A^{**}$, and show that it’s a left inverse of f .

$$g[] = []$$

$$g(a : s) = \begin{cases} [[a]] & \text{if } a \neq c \text{ and } s = [] \\ (a : t) : j & \text{if } a \neq c \text{ and } gs = t : j \\ [] : gs & \text{if } a = c \end{cases}$$

We then need to show that $g(f(j)) = j$ for all $j \in A^{**}$.

Strings over a countable alphabet

Note that whenever $A \lesssim B$, $A^* \lesssim B^*$, since any injection $f : A \rightarrow B$ can be lifted to an injection $f^* : A^* \rightarrow B^*$ by recursively defining

$$f^*[] = []$$

$$f^*(a : s) = fa : f^*s$$

So by the above, when A is finite and nonempty, and $B = A \cup \{c\}$ for some $c \notin A$, we have:

$$\mathbb{N}^* \sim A^{**} \sim B^* \sim \mathbb{N}$$

So we can conclude that whenever a set A is countable, A^* is countable too.

An easy corollary

Obviously $\mathbb{N} \times \mathbb{N} \lesssim \mathbb{N}^*$, since $[\langle n, m \rangle \rightarrow [n, m]]$ is injective.

And obviously $\mathbb{N} \lesssim \mathbb{N} \times \mathbb{N}$, since $[n \rightarrow \langle n, 0 \rangle]$ is injective.

So (by Schröder-Bernstein) we have $\mathbb{N} \sim \mathbb{N} \times \mathbb{N}$: there are as many ordered pairs of naturals as there are naturals.

Bijection $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$

The particular bijections we get by following the proof above are a bit wacky. But there are much nicer bijections, e.g. the one diagrammed here:

	0	1	2	3	4	...
0	0	1	3	6	10	
1	2	4	7	11		
2	5	8	12			
3	9	13				
4	14					
$\vdots$						

This is

$$f\langle n, m \rangle = \frac{(n+m)(n+m+1)}{2} + m$$

Countability facts

So we have analogues for countability for two of our facts about finitude.

Countability Fact 1

If A and B are countable, $A \cup B$ is countable.

Proof: suppose that $f : A \rightarrow \mathbb{N}$ and $g : B \rightarrow \mathbb{N}$ are injections. Then so is $h : A \cup B \rightarrow \mathbb{N}$ defined by

$$hx = \begin{cases} 2fx & \text{if } x \in A \\ 2gx + 1 & \text{if } x \in B \setminus A \end{cases}$$

Countability Fact 2

If A and B are countable, $A \times B$ is countable.

Proof: In general, if $A \lesssim X$ and $B \lesssim Y$ then $A \times B \lesssim X \times Y$.

We do *not* have an analogue of Finitude Fact 4: by Cantor's theorem, the powerset of a countably infinite set is *not* countably infinite.

We do however have a restricted version of this:

Countability Fact 4

If A is countable, $\mathcal{P}_{\text{fin}}A$ is countable.

Proof: We can injectively map sets in $\mathcal{P}_{\text{fin}}A$ to lists in A^* by ordering the elements according to some fixed $f : A \rightarrow \mathbb{N}$ (using the least number theorem).

Countability facts

We also *not* have an analogue of Finitude Fact 5. For example, although $\{0, 1\}$ and $\mathbb{N}$ are both countable, $\{0, 1\}^{\mathbb{N}}$ is not countable, since there is a bijection between it and $\mathcal{P}\mathbb{N}$.

We do however have the following restricted version:

Countability Fact 5

If A is countable and B is *finite*, A^B is countable.

Proof: by induction on the size of B .

Base step: $A^{\emptyset}$ has size 1.

Inductive step: there is a bijection from $A^{\text{pred } n} \times A$ to $A^{\text{pred suc } n}$, namely $[\langle f, a \rangle \mapsto f \cup \{\langle n, a \rangle\}]$.

Countable union of countable sets

It turns out that we also have a direct analogue of Finitude Fact 3:

Countability Fact 3

If $\mathbf{V}$ is countable and every element of $\mathbf{V}$ is countable, then $\bigcup V$ is countable.

Intuition: suppose for simplicity that no two elements of $\mathbf{V}$ overlap. Let f be an injection from $\mathbf{V}$ to $\mathbb{N}$, and for each $X \in \mathbf{V}$ let g_X be an injection from X to $\mathbb{N}$. Then the function h that maps each $y \in X$ to $\langle fX, g_X y \rangle$ is an injection from $\bigcup V$ to $\mathbb{N} \times \mathbb{N}$.

Making this proof rigorous turns out to involve an appeal to the Axiom of Choice.